

Development, simulation & implementation of precoding techniques for satellite links

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Ghent, May 2026

Wannes Baes

Acknowledgements

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by

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Ghent University

Faculty of Engineering and Architecture

Department of Telecommunications and Information Processing

Abstract — Laboris magna irure quis sit et veniam voluptate duis elit esse do excepteur. Officia laboris adipisicing laboris duis ut cillum aliquip. Do veniam aute non do cillum sit pariatur consectetur adipisicing qui. Lorem Lorem veniam minim exercitation cupidatat tempor do aute dolore. Nostrud adipisicing tempor quis qui fugiat sint quis ut aute eiusmod Lorem enim veniam. Elit labore Lorem do proident amet anim exercitation culpa enim cillum qui. Aliquip reprehenderit commodo labore minim laborum mollit ut veniam in irure. Minim sit in non aliqua ut elit veniam irure pariatur non velit consequat. Cupidatat veniam occaecat culpa eu irure aute consectetur dolore culpa deserunt quis officia do. Aliqua aliqua eu velit excepteur culpa ea dolore occaecat. Ea est cillum nulla ea cupidatat esse adipisicing ut. Cillum amet esse ex ex enim elit excepteur ullamco in qui aliqua Lorem veniam. Sunt consectetur aliquip commodo exercitation pariatur id tempor adipisicing aliqua. Fugiat cillum ea irure laborum consectetur ipsum. Excepteur laboris eu veniam ex ad consequat esse id ullamco ea cupidatat ad nisi. Aute nisi velit excepteur adipisicing dolore pariatur sunt. Dolore mollit cupidatat esse sint ipsum est nulla nisi eu cupidatat excepteur sint cillum. Culpa irure magna culpa eiusmod ut nisi ipsum veniam elit est occaecat veniam. Sunt adipisicing eu minim qui ea laboris ullamco ut. Aliquip occaecat in occaecat ut tempor dolore sunt consequat qui cillum reprehenderit. Voluptate commodo cupidatat ullamco sint aliquip veniam consequat do. Nisi velit excepteur aute laboris aliqua fugiat minim occaecat quis non incididunt.

Keywords — Laboris magna, Laboris magna, Laboris magna, Laboris magna

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$$a + b = \gamma \quad (1)$$

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- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [1].

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TABLE I
TABLE TYPE STYLES

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Fig. 1. Example of a figure caption.

them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

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- [1] A. I. Perez-Neira, M. A. Vazquez, M. B. Shankar, S. Maleki, and S. Chatzinotas, “Signal processing for high-throughput satellites: Challenges in new interference-limited scenarios,” *IEEE Signal Processing Magazine*, vol. 36, no. 4, pp. 112–131, 2019.

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Symbols and Notations

Abbreviations

BS	base station
CS	circularly symmetric
EVD	eigenvalue decomposition
IBR	information bit rate
i.i.d.	independent and identically distributed
KKT	Karush-Kuhn-Tucker
MI	mutual information
MIMO	multiple-input multiple-output
MSE	mean squared error
MU	multi-user
pdf	probability density function
QoS	Quality-of-Service
RX	receiver
SISO	single-input single-output
SNR	signal-to-noise ratio
SR	sum rate
SU	single-user
SVD	singular value decomposition
TX	transmitter
UT	user terminal
WMSE	weighted mean squared error
WMMSE	weighted minimum mean squared error

WSR weighted sum rate

Mathematical Notations

TO DO

System Parameters

N_T Number of transmit antennas at the BS

N_R Number of receive antennas at each UT

N_S Number of data streams per UT

K Number of UTs

Symbols

TO DO

Chapter 1

Introduction

Part I

Background Material

Chapter 2

Communication Systems

This chapter establishes the system models and information-theoretic framework that underlie the analysis presented throughout this work. We begin by describing the general architecture of a digital communication system and then specialize to the system model adopted in this work. Finally, we review the fundamental information-theoretic concepts required to analyze the performance of these systems.

2.1 Digital Communication Systems

A general digital communication system transfers digital information from a source to a destination over a physical transmission medium, referred to as the channel. The corresponding block diagram is shown in Fig. 2.1.

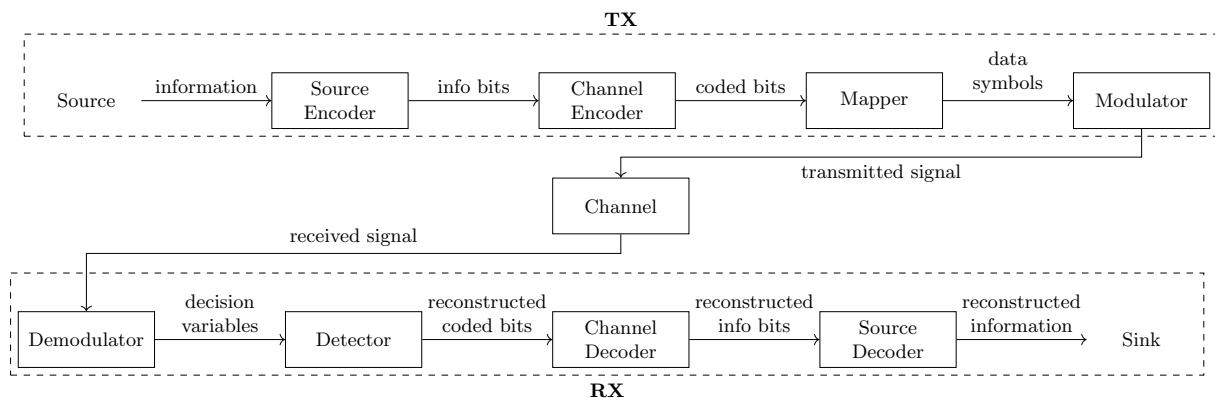


Figure 2.1: Block diagram of a general digital communication system.

At the transmitter (TX), the information sequence is first processed by a source encoder to remove redundancy, followed by a channel encoder that introduces controlled redundancy to enable error detection. The resulting coded bit sequence is partitioned into blocks of m_c bits, each of which is mapped onto a complex-valued symbol drawn from a constellation $\mathcal{X}$ of size 2^{m_c} . The modulator then converts the symbol sequence into a baseband signal suitable for transmission.

During transmission, the channel distorts the signal due to attenuation, multipath propagation, and additive noise, such that the received signal differs from the transmitted signal.

At the receiver (RX), the demodulator converts the received signal into a sequence of decision variables. These are processed by the detector to obtain estimates of the originally transmitted symbols, which are then mapped back to coded bits. Finally, the channel and source decoders reconstruct the original information sequence.

In this work, we consider a simplified version of this general architecture in order to focus on the aspects most relevant to our study. Source coding and decoding are omitted, and the information is assumed to be available as a sequence of independent and identically distributed (i.i.d.) bits without prior compression or digitization. Channel coding and decoding are also not considered, such that the analysis is restricted to uncoded transmission.

The transmission medium is assumed to be a wireless channel affected by fading and additive white Gaussian noise. Furthermore, the channel is considered to be frequency-flat, meaning that its response remains constant over the entire signal bandwidth. The specific channel models used in this work are introduced in the relevant chapters.

2.2 MIMO Communication Systems

When both the TX and the RX are equipped with multiple antennas, the system is referred to as a multiple-input multiple-output (MIMO) system.

Compared to a single-antenna link, the multiple antennas provide additional spatial degrees of freedom that can be exploited to increase data throughput, improve reliability, or achieve a trade-off between both. Spatial multiplexing techniques enable significant capacity gains by transmitting multiple independent data streams simultaneously over different antennas. A well-known example is the BLAST architecture [1]. Alternatively, spatial diversity can be achieved through space-time coding schemes, which distribute symbols across both space and time to improve robustness against fading. A classical example is the Alamouti scheme [2]. Throughout this work, the focus is on spatial multiplexing and, in particular, on how the simultaneous transmission of multiple data streams over a shared channel can be made most efficient through precoding.

2.2.1 SU-MIMO System Model

In the point-to-point case, where a single TX with N_T antennas communicates with a single RX with N_R antennas, the system is called a single-user (SU) MIMO system. A block diagram is shown in Fig. 2.2.

The discrete-time complex baseband input-output relation of a SU-MIMO Gaussian channel is characterized by

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (2.1)$$

where $\mathbf{x} \in \mathbb{C}^{N_T}$ is the transmitted signal vector, $\mathbf{y} \in \mathbb{C}^{N_R}$ is the received signal vector, $\mathbf{H} \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix, and $\mathbf{n} \in \mathbb{C}^{N_R}$ is the additive noise vector. The (n_r, n_t) -th entry of $\mathbf{H}$ represents the complex channel coefficient between transmit antenna n_t and receive

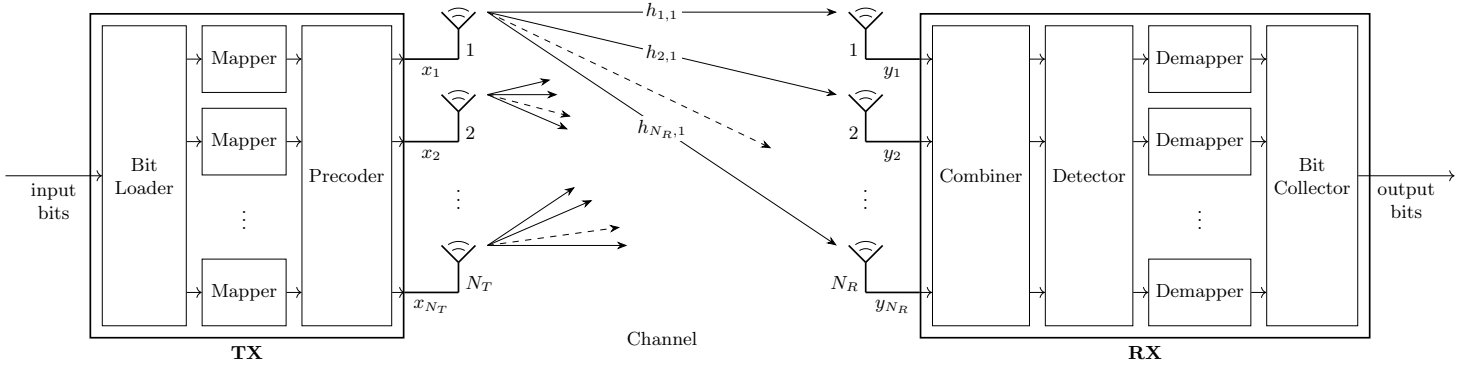


Figure 2.2: Block diagram of a SU-MIMO system.

antenna n_r . The noise vector is circularly symmetric (CS) complex Gaussian distributed, i.e. $\mathbf{n} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, N_0 \mathbf{I}_{N_R})$, such that its entries are spatially independent, each with an expected power of N_0 . The time index is omitted for brevity.

At the TX, the information bits are divided into N_S streams and mapped onto data symbols, which are collected in the vector $\mathbf{a} \in \mathbb{C}^{N_S \times 1}$. Then, the transmitted signal is obtained by mapping the N_S data symbol streams onto the N_T transmit antennas according to a linear precoding scheme, represented by a matrix $\mathbf{F} \in \mathbb{C}^{N_T \times N_S}$. The transmitted signal can therefore be expressed as

$$\mathbf{x} = \mathbf{F} \mathbf{a}. \quad (2.2)$$

The data symbols are assumed to be zero-mean, mutually uncorrelated, and normalized, such that $\mathbb{E}[\mathbf{a}\mathbf{a}^H] = \mathbf{I}_{N_S}$. Throughout this work, the power allocation is assumed to be incorporated into the precoding matrix, unless explicitly stated otherwise. Under this convention, the total transmit power is constrained by (see Appendix B)

$$\mathbb{E}[\|\mathbf{x}\|^2] = \text{tr}(\mathbf{F}^H \mathbf{F}) \leq P_t. \quad (2.3)$$

At the RX, the N_R received signals are linearly combined to obtain the decision-variable vector $\mathbf{z} \in \mathbb{C}^{N_S \times 1}$. The combiner is represented by a matrix $\mathbf{G} \in \mathbb{C}^{N_S \times N_R}$. The decision variables can therefore be expressed as

$$\mathbf{z} = \mathbf{G} \mathbf{y}. \quad (2.4)$$

Based on $\mathbf{z}$, the detector forms estimates of the transmitted data symbols, which are then mapped back to bits.

Substituting (2.2) into (2.1) and applying (2.4) yields the relation between the transmitted data symbols and the decision variables:

$$\mathbf{z} = (\mathbf{G} \mathbf{H} \mathbf{F}) \mathbf{a} + \mathbf{G} \mathbf{n}, \quad (2.5a)$$

$$= \mathbf{T} \mathbf{a} + \mathbf{n}_{\mathbf{G}}, \quad (2.5b)$$

where $\mathbf{T} \in \mathbb{C}^{N_S \times N_S}$ is the MIMO transfer matrix and $\mathbf{n}_{\mathbf{G}}$ represents spatially correlated Gaussian noise. It is evident that, whenever the transfer matrix is not diagonal, the data symbols of different

streams interfere with each other. The design of the precoding and combining matrices is therefore essential to suppress this inter-stream interference and improve overall system performance.

2.2.2 MU-MIMO System Model

When multiple receivers are served simultaneously by a single multi-antenna transmitter, the system is referred to as a multi-user (MU) MIMO system. In this context, the transmitter is called the base station (BS) and the individual receivers are referred to as user terminals (UTs). A block diagram is shown in Fig. 2.3.

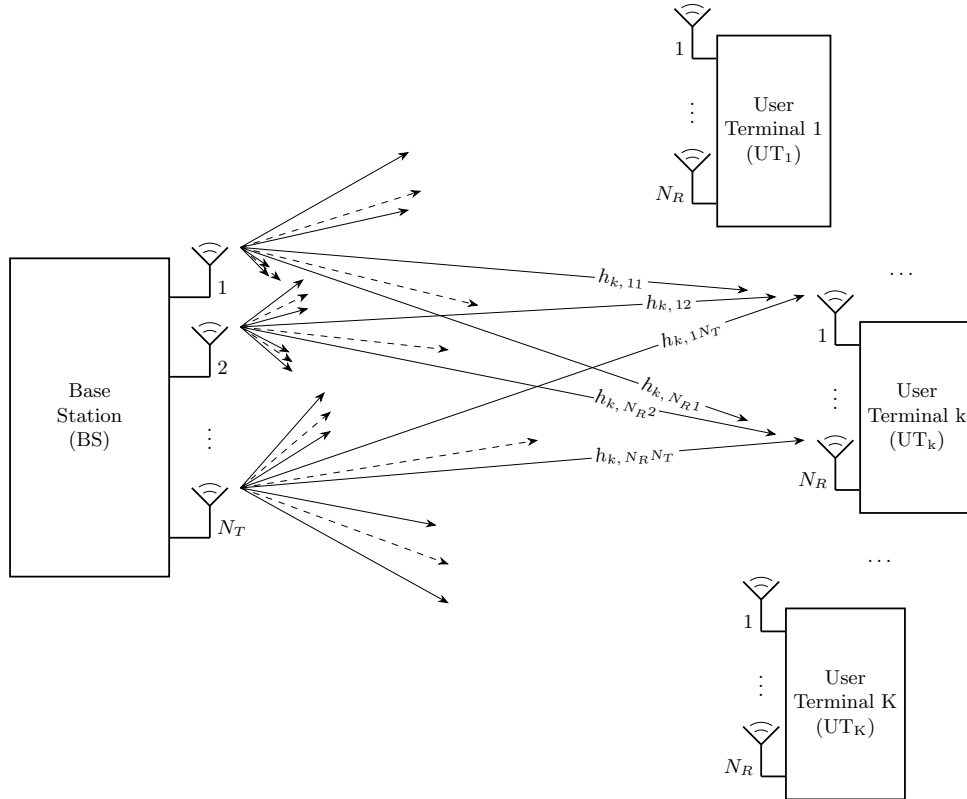


Figure 2.3: Block diagram of a MU-MIMO system.

Throughout this work, the BS is assumed to be equipped with more antennas than all UTs combined, and each UT is assumed to receive at most as many data streams as it has receive antennas. For notational convenience, all UTs are assumed to receive the same number of data streams, denoted by N_S . In practical scenarios where this does not hold, N_S is defined as the maximum number of data streams intended for any single UT. The data symbol vectors intended for a UT receiving fewer streams are padded with zeros.

Analogous to the SU-MIMO case, the communication between the BS and a certain UT_k at a particular time instant is characterized by

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{x} + \mathbf{n}_k, \quad k = 1, \dots, K, \quad (2.6)$$

where $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix that describes the link between the BS and UT_k .

The transmitted signal vector $\mathbf{x} \in \mathbb{C}^{N_T}$ is now a function of the data symbol vectors intended for different UTs. The BS assigns a dedicated precoding scheme $\mathbf{F}_k \in \mathbb{C}^{N_T \times N_S}$ to the data symbol vector $\mathbf{a}_k$ intended for UT_k, and transmits the superposition of all precoded streams:

$$\mathbf{x} = \sum_{k=1}^K \mathbf{F}_k \mathbf{a}_k. \quad (2.7)$$

Since the data symbols intended for different UTs are assumed to be uncorrelated, the total transmit power constraint can now be formulated as (see Appendix B)

$$\mathbb{E}[\|\mathbf{x}\|^2] = \sum_{k=1}^K \text{tr}(\mathbf{F}_k^H \mathbf{F}_k) \leq P_t. \quad (2.8)$$

Applying the combining scheme $\mathbf{G}_k \in \mathbb{C}^{N_S \times N_R}$ at UT_k to the received signal vector and substituting (2.7) into (2.6) yields

$$\mathbf{z}_k = \underbrace{(\mathbf{G}_k \mathbf{H}_k \mathbf{F}_k)}_{\text{desired signal}} \mathbf{a}_k + \underbrace{\sum_{k'=1, k' \neq k}^K (\mathbf{G}_k \mathbf{H}_k \mathbf{F}_{k'}) \mathbf{a}_{k'}}_{\text{inter-user interference}} + \underbrace{\mathbf{G}_k \mathbf{n}_k}_{\text{noise}}. \quad (2.9)$$

The key difference from the SU-MIMO scenario is the presence of inter-user interference. Each UT receives not only its desired signal but also the signals intended for all other UTs. Suppressing this interference through precoder design at the BS is an important challenge in MU-MIMO systems.

The overall input-output relation of the MU-MIMO Gaussian channel can be written compactly as

$$\mathbf{z} = (\mathbf{G} \mathbf{H} \mathbf{F}) \mathbf{a} + \mathbf{G} \mathbf{n}, \quad \text{with} \quad (2.10)$$

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_K \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \mathbf{G}_1 & & \\ & \ddots & \\ & & \mathbf{G}_K \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_1 \\ \vdots \\ \mathbf{H}_K \end{bmatrix}, \quad \mathbf{F} = [\mathbf{F}_1 \quad \dots \quad \mathbf{F}_K], \quad \mathbf{a} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_K \end{bmatrix}.$$

where $\mathbf{G} \in \mathbb{C}^{KN_S \times KN_R}$ denotes the compound combining matrix, $\mathbf{H} \in \mathbb{C}^{KN_R \times N_T}$ the compound channel matrix, and $\mathbf{F} \in \mathbb{C}^{N_T \times KN_S}$ the compound precoding matrix.

2.3 Information Theory Concepts

In this section, we introduce the information-theoretic concepts needed for the analysis of MIMO communication systems. We first discuss mutual information, then define the achievable rate and capacity for SU-MIMO channels, and finally extend these concepts to the MU scenario.

2.3.1 Mutual Information

Consider a single use of a SU-MIMO Gaussian channel, as described by (2.1). The transmitter generates a random vector $\mathbf{x} \in \mathbb{C}^{N_T}$ according to the input distribution with probability density function (pdf) $p(\mathbf{x})$, while the receiver observes the random vector $\mathbf{y} \in \mathbb{C}^{N_R}$. The relation between the channel input $\mathbf{x}$ and the channel output $\mathbf{y}$ is probabilistic and is fully characterized by the channel law, described by the conditional pdf $p(\mathbf{y} | \mathbf{x})$.

The number of information bits transmitted per channel use is referred to as the information bit rate (IBR). The mutual information (MI) between $\mathbf{x}$ and $\mathbf{y}$ quantifies how much information the received vector $\mathbf{y}$ provides about the transmitted vector $\mathbf{x}$. In other words, it represents the average reduction in the uncertainty about $\mathbf{x}$ obtained from observing $\mathbf{y}$. It is given by

$$I(\mathbf{x}; \mathbf{y}) = \mathbb{E} \left[\log_2 \left(\frac{p(\mathbf{y} | \mathbf{x})}{p(\mathbf{y})} \right) \right], \quad (2.11)$$

where the expectation is taken with respect to the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ and $p(\mathbf{y})$ is the marginal distribution of $\mathbf{y}$.

According to Shannon's channel coding theorem [3], reliable communication is possible if the transmission IBR is strictly smaller than the MI between $\mathbf{x}$ and $\mathbf{y}$. Here, reliable communication means that the decoding error probability can be made arbitrarily small by using sufficiently long channel codes.

2.3.2 Achievable Rate and Capacity

For a given input distribution $p(\mathbf{x})$ and channel law $p(\mathbf{y} | \mathbf{x})$, any IBR r smaller than the MI is said to be achievable. The MI is thus a supremum (least upper bound) on the achievable rate.

The capacity C of a SU-MIMO channel is the largest achievable rate. Equivalently, it is the highest spectral efficiency for which reliable communication is possible. It is defined as the maximum MI between $\mathbf{x}$ and $\mathbf{y}$ over all possible input distributions $p(\mathbf{x})$:

$$C = \max_{p(\mathbf{x})} I(\mathbf{x}; \mathbf{y}). \quad (2.12)$$

Typically, the maximization is subject to the transmit power constraint, as formulated in (2.3).

In [4], the capacity for a SU-MIMO Gaussian communication system is derived. First, it is shown that under the total transmit power constraint, the MI is maximized in case of Gaussian signaling, i.e. the channel input $\mathbf{x}$ is CS complex Gaussian distributed. Then, the mutual information between $\mathbf{x}$ and $\mathbf{y}$ for a given covariance matrix $\mathbf{C}_x$ is obtained as

$$I(\mathbf{x}; \mathbf{y}) = \log_2 \left(\det \left(\mathbf{I}_{N_R} + \mathbf{C}_n^{-1} \mathbf{H} \mathbf{C}_x \mathbf{H}^H \right) \right). \quad (2.13)$$

The capacity then results from maximizing (2.13) over all positive semidefinite covariance matrices $\mathbf{C}_x$ that satisfy the transmit power constraint:

$$\begin{aligned} C &= \max_{\mathbf{C}_x} \log_2 (\det (\mathbf{I}_{N_R} + \mathbf{C}_n^{-1} \mathbf{H} \mathbf{C}_x \mathbf{H}^H)) \\ \text{s.t. } &\mathbf{C}_x \succeq 0, \quad \text{tr}(\mathbf{C}_x) \leq P_t. \end{aligned} \quad (2.14)$$

2.3.3 Rate Region and Sum Capacity

In contrast to the SU scenario, where the IBR is a scalar r , a MU-MIMO channel is characterized by a rate tuple $(r_1, \dots, r_K)$, where each element r_k denotes the IBR of a single UT. The rate region $\mathcal{R}$ is the set of all rate tuples whose individual IBRs are simultaneously achievable under the given system constraints. The rate region is a convex set, since any convex combination of achievable rate tuples is itself achievable through time-sharing between the corresponding transmission strategies. An example rate region for $K = 2$ users is shown in Fig. 2.4.

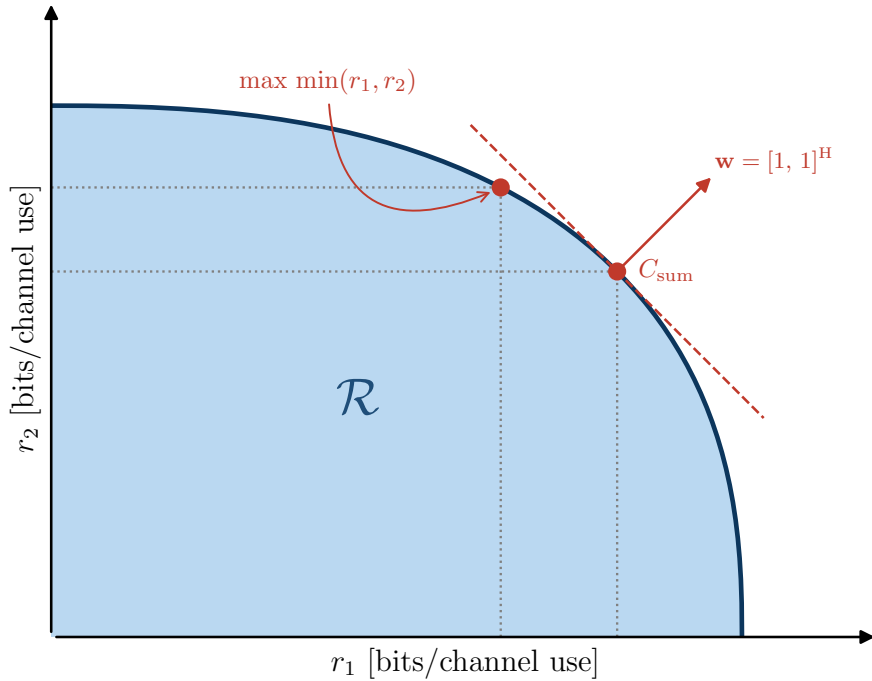


Figure 2.4: Example of a rate region $\mathcal{R}$ for a MU-MIMO system with $K = 2$ users.

The sum capacity point is the boundary point whose outward normal is the all-ones vector. The max-min point is given by the intersection of the boundary with the line $r_1 = r_2$. It corresponds to a fair operating point where both users achieve the same IBR.

The rate region captures the trade-offs inherent to multi-user transmission: increasing the IBR of one user may reduce the rates achievable by others due to interference and shared resources. It is therefore desirable to operate on the boundary of the rate region, where no user's rate can be increased without reducing that of another.

The sum capacity C_{sum} is the maximum achievable sum rate (SR), i.e., the sum of all individual IBRs:

$$C_{\text{sum}} = \max_{(r_1, \dots, r_K) \in \mathcal{R}} \sum_{k=1}^K r_k. \quad (2.15)$$

More generally, we define the weighted sum rate (WSR) as

$$\text{WSR} = \sum_{k=1}^K w_k r_k, \quad (2.16)$$

where $w_k \geq 0$ is the weight assigned to UT_k . The full boundary of the rate region can then be traced by maximizing the WSR over all achievable rate tuples for varying weight vectors $\mathbf{w} = (w_1, \dots, w_K)$. The sum capacity corresponds to the special case where all weights are equal to one. The maximization in both cases is typically subject to the total transmit power constraint (2.8).

In [5, 6, 7], the sum capacity of the MU-MIMO Gaussian channel is derived by exploiting the uplink–downlink duality: the downlink broadcast channel and the uplink multiple-access channel yield the same rate region under the same total transmit power constraint. It therefore suffices to solve the uplink problem, which is more tractable.

In the uplink channel, all K UTs transmit simultaneously, with UT_k sending a vector $\mathbf{x}_k^{\text{up}} \in \mathbb{C}^{N_R}$ to the BS. The MI between the transmitted vectors $\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}$ and the received vector $\mathbf{y}$, given input covariance matrices $\mathbf{C}_{\mathbf{x},k}$, is defined as

$$I(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \mathbb{E} \left[\log_2 \left(\frac{p(\mathbf{y} | \mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}})}{p(\mathbf{y})} \right) \right], \quad (2.17)$$

where the expectation is taken with respect to the joint pdf of $(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}})$ and $\mathbf{y}$, and $p(\mathbf{y})$ is the marginal pdf of $\mathbf{y}$. It is further obtained as

$$I(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \log_2 \left(\det \left(\mathbf{I}_{N_T} + \mathbf{C}_n^{-1} \sum_{k=1}^K \mathbf{H}_k \mathbf{C}_{\mathbf{x},k}^{\text{up}} \mathbf{H}_k^H \right) \right). \quad (2.18)$$

The uplink SR is achievable if and only if the sum of the IBRs of all UTs is strictly smaller than the MI as given in (2.18). Therefore, the sum capacity of the downlink channel is obtained by maximizing (2.18) over all positive semidefinite covariance matrices satisfying the total transmit power constraint and applying the duality result:

$$\begin{aligned} C_{\text{sum}} &= \max_{\mathbf{C}_{\mathbf{x},1}^{\text{up}}, \dots, \mathbf{C}_{\mathbf{x},K}^{\text{up}}} \log_2 \left(\det \left(\mathbf{I}_{N_T} + \mathbf{C}_n^{-1} \sum_{k=1}^K \mathbf{H}_k \mathbf{C}_{\mathbf{x},k}^{\text{up}} \mathbf{H}_k^H \right) \right) \\ \text{s.t.} \quad &\sum_{k=1}^K \text{tr}(\mathbf{C}_{\mathbf{x},k}^{\text{up}}) \leq P_t \quad \text{and} \quad \mathbf{C}_{\mathbf{x},k}^{\text{up}} \succeq 0 \quad \forall k = 1, \dots, K. \end{aligned} \quad (2.19)$$

Chapter 3

Precoding for SU-MIMO systems

3.1 Capacity Achieving Precoding

Consider the SU-MIMO system model introduced in Section 2.2.1 together with the associated capacity optimization problem in (2.14). We first derive a more explicit expression for the channel capacity and then show that this capacity can be achieved by means of linear precoding. Parts of this section follow a structure similar to that in Chapter 3 of [8].

3.1.1 SU-MIMO Capacity

The singular value decomposition (SVD) [9] of the channel matrix yields

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H, \quad (3.1)$$

where $\mathbf{U} \in \mathbb{C}^{N_R \times N_R}$ and $\mathbf{V} \in \mathbb{C}^{N_T \times N_T}$ are unitary matrices, and $\mathbf{\Sigma} \in \mathbb{R}^{N_R \times N_T}$ is a rectangular diagonal matrix which contains the singular values of $\mathbf{H}$ on its main diagonal in decreasing order. The columns of $\mathbf{U}$ and $\mathbf{V}$ contain the left and right singular vectors of $\mathbf{H}$, respectively.

Now define the transformed variables $\tilde{\mathbf{y}} = \mathbf{U}^H \mathbf{y}$, $\tilde{\mathbf{x}} = \mathbf{V}^H \mathbf{x}$, and $\tilde{\mathbf{n}} = \mathbf{U}^H \mathbf{n}$. Since multiplication by a unitary matrix preserves the distribution of CS complex Gaussian noise, $\tilde{\mathbf{n}}$ has the same distribution as $\mathbf{n}$. Moreover, because $\mathbf{U}$ and $\mathbf{V}$ are unitary, the transformations between $(\mathbf{x}, \mathbf{y})$ and $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ are bijective. The original channel from (2.1) can therefore be represented equivalently as

$$\tilde{\mathbf{y}} = \mathbf{\Sigma} \tilde{\mathbf{x}} + \tilde{\mathbf{n}}, \quad (3.2)$$

which is obtained by applying the SVD of the channel matrix and performing a change of variables (see Appendix B). The channel in (3.2) is equivalent to the original channel in the sense that the transformation preserves the mutual information between input and output. Consequently, both channels have the same capacity.

The channel matrix $\mathbf{H}$ has exactly as many non-zero singular values $\{\sigma_i\}$ as its rank, denoted as R_H . Written component-wise, the equivalent channel becomes

$$\tilde{y}_i = \sigma_i \tilde{x}_i + \tilde{n}_i \quad \text{for } i = 1, \dots, R_H, \quad (3.3)$$

with $\tilde{y}_i = \tilde{n}_i$ for $R_H < i \leq N_R$. The equivalent channel decomposes into R_H independent parallel scalar Gaussian subchannels, each with its own channel gain σ_i . These subchannels and their gains are referred to as eigen subchannels and eigenmodes, respectively. Since the components $\tilde{x}_i$ for $R_H < i \leq N_T$ have no effect on the output, they carry no information and can be set to zero.

For each eigen subchannel, the mutual information (MI) is maximised by a zero-mean, CS complex Gaussian input [4]. Under this choice, the optimization problem (2.14) simplifies to

$$\begin{aligned} C &= \max_{\{\mathbb{E}[|\tilde{x}_i|^2]\}} \sum_{i=1}^{R_H} \log_2 \left(1 + \mathbb{E}[|\tilde{x}_i|^2] \frac{\sigma_i^2}{N_0} \right) \\ \text{s.t. } &\sum_{i=1}^{R_H} \mathbb{E}[|\tilde{x}_i|^2] \leq P_t. \end{aligned} \quad (3.4)$$

This maximization problem is solved using the Lagrange multiplier method (see Appendix A). The variances of the channel input need to be chosen according to the waterfilling solution [10]:

$$\mathbb{E}[|\tilde{x}_i|^2] = \left(\mu_w - \frac{N_0}{\sigma_i^2} \right)_+, \quad (3.5)$$

where the water level μ_w is found iteratively to satisfy the transmit power constraint with equality.

The resulting capacity is given by

$$C = \sum_{i=1}^{R_H} \log_2 \left(\max(1, \mu_w \frac{\sigma_i^2}{N_0}) \right). \quad (3.6)$$

3.1.2 Optimal Precoder Design

In this section, we design a linear precoder and combiner that together achieve the capacity of the SU-MIMO channel. We assume that the data symbol vector is CS complex Gaussian distributed and spatially uncorrelated, and restrict ourselves to the case where the number of data streams is not fixed in advance, but can be chosen freely as part of the design.

For an arbitrary precoding matrix $\mathbf{F}$ and combining matrix $\mathbf{G}$, the data processing inequality [11] yields $I(\mathbf{x}; \mathbf{y}) \geq I(\mathbf{a}; \mathbf{z})$, where the MI between the transmitted data symbols $\mathbf{a}$ and the decision variables $\mathbf{z}$ is given by:

$$I(\mathbf{a}; \mathbf{z}) = \log_2 \left(\det(\mathbf{I}_{N_s} + (\mathbf{G} \mathbf{C}_n \mathbf{G}^H)^{-1} (\mathbf{G} \mathbf{H} \mathbf{F}) \mathbf{C}_a (\mathbf{G} \mathbf{H} \mathbf{F})^H) \right) \quad (3.7a)$$

$$= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} (\mathbf{G} \mathbf{G}^H)^{-1} (\mathbf{G} \mathbf{H} \mathbf{F}) (\mathbf{G} \mathbf{H} \mathbf{F})^H \right) \right). \quad (3.7b)$$

Expression (3.7a) follows directly from (2.13) and the structural similarity between (2.1) and (2.5). Expression (3.7b) then follows from (3.7a) upon substituting $\mathbf{C}_a = \mathbf{I}_{N_s}$ and $\mathbf{C}_n = N_0 \mathbf{I}_{N_R}$.

Motivated by the capacity derivation in the previous subsection, we choose the columns of $\mathbf{F}$ and the rows of $\mathbf{G}$ as the R_H most dominant right and left singular vectors of $\mathbf{H}$, respectively. These are precisely the singular vectors associated with the non-zero singular values of $\mathbf{H}$. The number of data streams is therefore chosen equal to the rank of the channel matrix. In addition, the precoder incorporates the power allocation through a diagonal matrix $\mathbf{P}$. More specifically, we set

$$\mathbf{F} = \mathbf{V}_{N_s} \sqrt{\mathbf{P}} \quad \text{and} \quad \mathbf{G} = \mathbf{U}_{N_s}^H. \quad (3.8)$$

It follows from (3.8) that $I(\mathbf{a}; \mathbf{z}) = I(\mathbf{x}; \mathbf{y})$ for an appropriate choice of the power allocation matrix $\mathbf{P}$. This result can be verified in a brute-force way by substituting the particular precoder and combiner matrices from (3.8) into (3.7) and replacing the channel matrix by its SVD (see Appendix B).

An alternative and more elegant approach consists in substituting the particular precoder and combiner into (2.5) and comparing the resulting expression with the equivalent channel given in (3.2). This yields

$$\mathbf{z} = \mathbf{U}_{N_s}^H \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H \mathbf{V}_{N_s} \sqrt{\mathbf{P}} \mathbf{a} + \mathbf{U}_{N_s}^H \mathbf{n} \quad (3.9a)$$

$$= \mathbf{\Sigma}_{N_s} \sqrt{\mathbf{P}} \mathbf{a} + \tilde{\mathbf{n}}, \quad (3.9b)$$

where $\mathbf{\Sigma}_{N_s} \in \mathbb{R}^{N_s \times N_s}$ contains the R_H non-zero singular values of $\mathbf{H}$ on its diagonal. Comparing (3.9) with (3.2), one can readily verify that $I(\mathbf{a}; \mathbf{z}) = I(\mathbf{x}; \mathbf{y})$, provided that the power allocated to each stream is set according to the waterfilling solution. Hence, transmission with the precoder and combiner from (3.8) achieves the capacity of the SU-MIMO channel.

Denoting the s th element of the decision variable vector $\mathbf{z}$ by z_s , one can write (3.9) component-wise as

$$z_s = \sigma_s \sqrt{p_s} a_s + \tilde{n}_s \quad \text{for } s = 1, \dots, N_s. \quad (3.10)$$

This expression shows that the precoder and combiner in (3.8) diagonalize the SU-MIMO transfer matrix from (2.5). The original SU-MIMO channel is decoupled into $N_s = R_H$ independent parallel scalar single-input single-output (SISO) Gaussian subchannels with gains σ_s , which coincides exactly with the decomposition of the equivalent channel in (3.3). A visualization of this decomposition is shown in Fig. 3.1.

As a result, the individual data streams become statistically independent. Detection therefore reduces to symbol-by-symbol processing rather than joint detection of the full decision-variable vector. This leads to a significant reduction in receiver complexity, especially for systems with a large number of antennas and data streams.

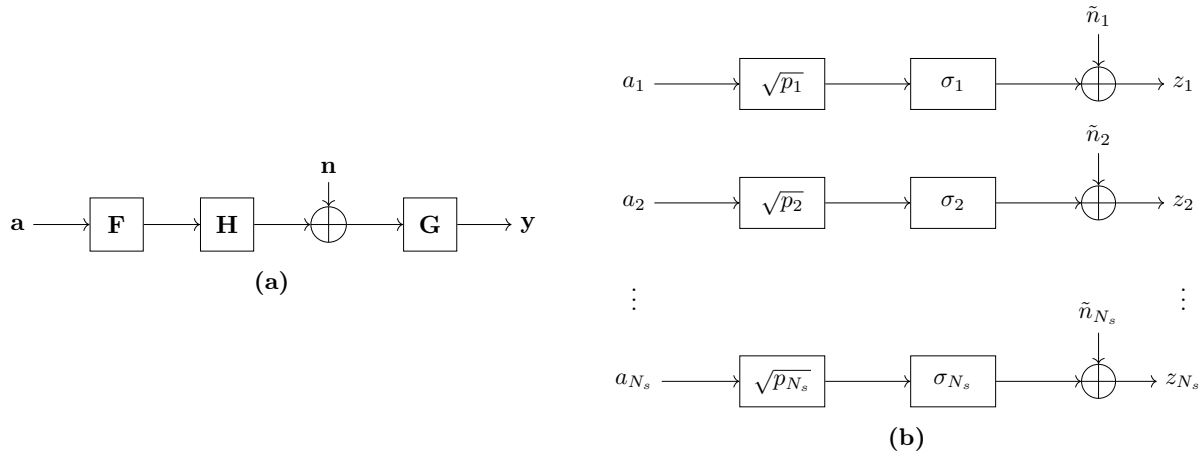


Figure 3.1: A block diagram of the original SU-MIMO channel (a) and its decomposition into parallel SISO Gaussian eigen subchannels (b) by means of the optimal precoder and combiner.

No general closed-form expression exists for the optimal water level μ_w , and hence for the resulting power allocation. Nevertheless, the optimal power allocation is fully characterized by

$$\sum_{i=1}^{N_s} p_i = P_t \quad \text{and} \quad p_i = \left(\mu_w - \frac{N_0}{\sigma_i^2} \right)_+ . \quad (3.11)$$

Since the total allocated power is a monotonically increasing function of the water level, the latter can be determined numerically by means of a bisection method. A more efficient approach exploits the geometric interpretation of the waterfilling solution. In the present SU-MIMO setting, each eigen subchannel can be represented as a container with unit width and floor level N_0/σ_i^2 , while the total transmit power is interpreted as water that is poured into these containers until a common water level is reached. The complete procedure is summarized in Appendix A.2. The structure of the optimal power allocation is illustrated in Fig. 3.2 for a SU-MIMO system with four transmit and four receive antennas operating at an signal-to-noise ratio (SNR) of 5 dB: eigen subchannels with stronger channel conditions, i.e. larger gains σ_i , receive more power than weaker ones.

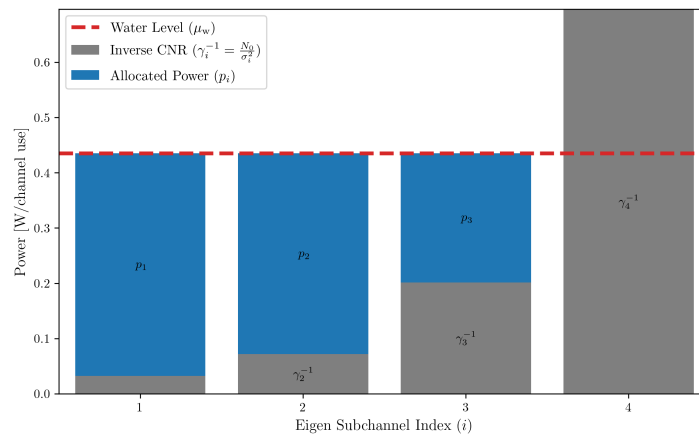


Figure 3.2: Optimal power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas and a total transmit power P_t of 1 W per channel use. The system operates at an SNR of 5 dB.

At extremely low SNR, the waterfilling solution corresponds to the eigenbeamforming power allocation strategy, which concentrates all available power on the strongest eigen subchannel. As a result, only a single data stream is transmitted in this SNR regime. This behavior is illustrated in Fig. 3.3a. Conversely, at an extremely high SNR, the optimal power allocation strategy reduces to the uniform power allocation, which distributes power equally across all eigen subchannels, as illustrated in Fig. 3.3b. It is worth noting that uniform power allocation does not require knowledge of the current channel matrix, in contrast to the general waterfilling solution.

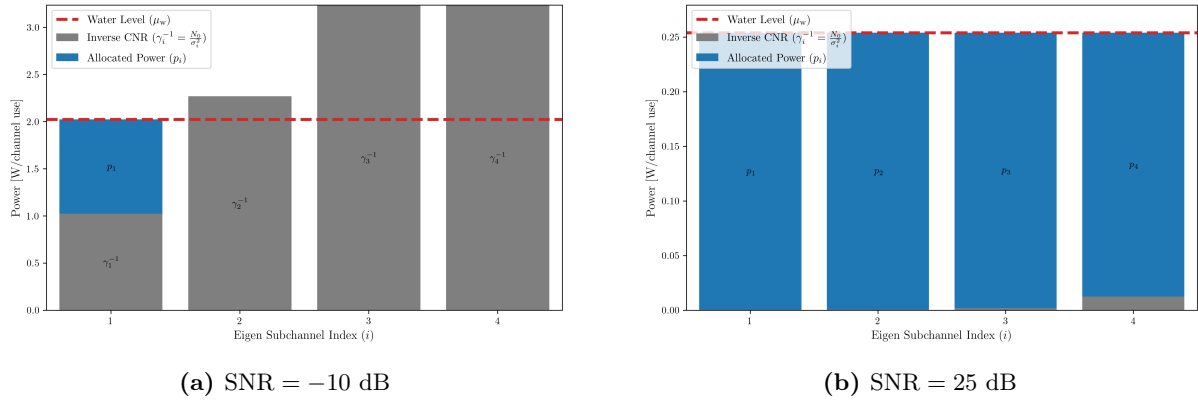


Figure 3.3: Waterfilling power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas at different SNR extremes. The maximum transmit power P_t is 1 W per channel use.

3.2 Joint Design for Linear Precoder and Combiner

In the previous section, a linear precoder and combiner that jointly decompose the SU-MIMO channel into independent parallel eigen subchannels were shown to achieve the channel capacity. In this section, we generalize this result by adopting the minimization of a weighted sum of symbol estimation errors as the design criterion. It will be shown that the optimum precoder and combiner under this criterion likewise decouple the channel into parallel eigen subchannels, regardless of the specific choice of error weights. By an appropriate selection of these weights, the framework specializes to the maximum information rate design, the quality-of-service design, and the minimum sum-of-errors design.

3.2.1 Generalized WMMSE Design

The objective is to jointly design the precoder $\mathbf{F}$ and the combiner $\mathbf{G}$ so as to minimize a weighted combination of symbol estimation errors. To this end, the estimation error vector $\mathbf{e} \in \mathbb{C}^{N_s}$ is defined as the difference between the transmitted data symbol vector $\mathbf{a}$ and the decision variable vector $\mathbf{z}$:

$$\mathbf{e} = \mathbf{a} - \mathbf{z} = \mathbf{a} - (\mathbf{G}\mathbf{H}\mathbf{F}\mathbf{a} + \mathbf{G}\mathbf{n}). \quad (3.12)$$

The mean squared error (MSE) matrix, also referred to as the error covariance matrix, is defined as

$$\text{MSE} = \mathbb{E}[\mathbf{e}\mathbf{e}^H] = (\mathbf{G}\mathbf{H}\mathbf{F} - \mathbf{I}_{N_s})(\mathbf{G}\mathbf{H}\mathbf{F} - \mathbf{I}_{N_s})^H + \mathbf{G}\mathbf{C}_n\mathbf{G}^H \quad (3.13)$$

where the last equality follows from the independence of the data symbols $\mathbf{a}$ and the noise vector $\mathbf{n}$, and the assumption that the data symbols are spatially uncorrelated and normalized.

The weighted mean squared error (WMSE) is defined as

$$\text{WMSE} = \mathbb{E} \left[\left\| \mathbf{W}_e^{1/2} \mathbf{e} \right\|^2 \right] = \mathbb{E} \left[\text{tr}(\mathbf{W}_e^{1/2} \mathbf{e} \mathbf{e}^H \mathbf{W}_e^{1/2}) \right] = \text{tr}(\mathbf{W}_e \mathbb{E}[\mathbf{e} \mathbf{e}^H]), \quad (3.14)$$

where $\mathbf{W}_e$ is a diagonal positive definite weight matrix that assigns a relative importance to the error of each data stream.

The weighted minimum mean squared error (WMMSE) optimization problem is then formulated as

$$\begin{aligned} \min_{\mathbf{F}, \mathbf{G}} \quad & \text{tr}(\mathbf{W}_e \mathbb{E}[\mathbf{e} \mathbf{e}^H]) \\ \text{s. t.} \quad & \text{tr}(\mathbf{F} \mathbf{F}^H) \leq P_t. \end{aligned} \quad (3.15)$$

The method of Langrange duality and the Karush-Kuhn-Tucker (KKT) conditions can be used to solve the optimization problem in (3.15). This leads to the following necessary optimality conditions [12]:

$$\mathbf{H} \mathbf{F} = \mathbf{H} \mathbf{F} (\mathbf{G} \mathbf{H} \mathbf{F})^H + \mathbf{C}_n \mathbf{G}^H \quad (3.16a)$$

$$\mathbf{W}_e \mathbf{G} \mathbf{H} = (\mathbf{G} \mathbf{H} \mathbf{F})^H \mathbf{W}_e \mathbf{G} \mathbf{H} + \mu \mathbf{F}^H \quad (3.16b)$$

where $\mu > 0$ is the Lagrange multiplier associated with the total transmit power constraint.

Solving these coupled matrix equations yields the optimum linear precoder and combiner. As shown in [12], they admit the following closed-form expressions:

$$\mathbf{F} = \mathbf{V} \Phi_f \quad (3.17a)$$

$$\mathbf{G} = \Phi_g \mathbf{V}^H \mathbf{H}^H \mathbf{C}_n^{-1} \quad (3.17b)$$

The matrices $\Phi_f, \Phi_g \in \mathbb{R}^{R_H \times R_H}$ are diagonal matrices with non-negative diagonal entries, and are given by

$$\Phi_f = (\mu^{-1/2} \mathbf{\Lambda}^{-1/2} \mathbf{W}_e^{1/2} - \mathbf{\Lambda}^{-1})_+^{1/2} \quad (3.18a)$$

$$\Phi_g = (\mu^{1/2} \mathbf{\Lambda}^{-1/2} \mathbf{W}_e^{-1/2} - \mu \mathbf{\Lambda}^{-1} \mathbf{W}_e^{-1})_+^{1/2} \mathbf{\Lambda}^{1/2} \quad (3.18b)$$

The matrices $\mathbf{V} \in \mathbb{C}^{N_T \times R_H}$ and $\mathbf{\Lambda} \in \mathbb{R}^{R_H \times R_H}$ are obtained from the eigenvalue decomposition (EVD) of the noise-whitened Gram matrix of $\mathbf{H}$:

$$\mathbf{H}^H \mathbf{C}_n^{-1} \mathbf{H} = \begin{bmatrix} \mathbf{V} & \bar{\mathbf{V}} \end{bmatrix} \begin{bmatrix} \mathbf{\Lambda} & \mathbf{0} \\ \mathbf{0} & \bar{\mathbf{\Lambda}} \end{bmatrix} \begin{bmatrix} \mathbf{V} & \bar{\mathbf{V}} \end{bmatrix}^H, \quad (3.19)$$

where the columns of $\mathbf{V}$ form an orthonormal basis for the range space of $(\mathbf{H}^H \mathbf{C}_n^{-1} \mathbf{H})$, and the columns of $\bar{\mathbf{V}}$ span its null space. The diagonal matrix $\mathbf{\Lambda}$ contains the R_H non-zero eigenvalues arranged in descending order, whereas $\bar{\mathbf{\Lambda}}$ contains the remaining zero eigenvalues. Note that, as

in Section 3.1.2, the number of data streams is set equal to the rank of the channel matrix, i.e. $N_s = R_H$. Moreover, the rank of the channel matrix is equal to the rank of $(\mathbf{H}^H \mathbf{C}_n^{-1} \mathbf{H})$ since $\mathbf{C}_n^{-1}$ is positive definite and therefore does not alter the rank of the product.

Now, it only lasts us to show how to obtain the Lagrange multiplier that satisfies the power constraint and also is such that the bracketed matrices in both (3.18a) and (3.18b) are positive-definite. Let $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_{R_H})$ and $\mathbf{W}_e = \text{diag}(w_1, \dots, w_{R_H})$. Suppose $k < R_H$ eigen subchannels are effectively used for transmission, i.e. they are allocated non-zero power. From (3.18a) and the total power constraint in (3.15), we have

$$\text{tr}(\mathbf{F}\mathbf{F}^H) = \text{tr}(\mathbf{\Phi}_f^2) = \mu^{-1/2} \sum_{i=1}^k \lambda_i^{-1/2} w_i^{1/2} - \sum_{i=1}^k \lambda_i^{-1} = P_t. \quad (3.20)$$

Therefore, we can obtain the following expression for the Lagrange multiplier μ :

$$\mu = \left(\frac{\sum_{i=1}^k \lambda_i^{-1/2} w_i^{1/2}}{P_t + \sum_{i=1}^k \lambda_i^{-1}} \right)^2. \quad (3.21)$$

Let $\rho_k = \lambda_k w_k$ and let them be ordered in decreasing manner. To compute $\mathbf{\Phi}_f$ optimally and ensure that it is positive-definite, we can compute μ according to (3.21) for each $k = 1, \dots, R_H$ iteratively, and select the largest k such that $\mu \leq \rho_k$ [12].

From (3.17a) and (3.17b), we have:

$$\mathbf{T} = \mathbf{G}\mathbf{H}\mathbf{F} = \mathbf{\Phi}_g \mathbf{\Lambda} \mathbf{\Phi}_f. \quad (3.22)$$

In other words, the optimum precoder and combiner diagonalize the SU-MIMO transfer matrix $\mathbf{T}$, and hence decouple the channel into parallel eigen subchannels, regardless of the specific choice of error weights. This result is similar to the capacity-achieving precoder and combiner from Section 3.1.2, who also decompose the channel matrix $\mathbf{H}$.

We can now design a linear precoder and combiner according to different design criteria, such as the maximize the information rate, minimize the sum of estimation errors, or meet certain quality-of-service requirements by choosing an appropriate error weight matrix $\mathbf{W}_e$. In the following, we will explore some of these design criteria in more detail.

3.2.2 Maximum Information Rate Design

It was already shown in Section 3.1.2 that the capacity of the SU-MIMO channel can be achieved by a linear precoder and combiner that diagonalize the transfer matrix and employ a waterfilling power allocation strategy. Rewriting (3.11) in matrix format and switching to the parameters of the generalized design, we obtain [12]

$$\mathbf{\Phi}_f = (\mu^{-1/2} \mathbf{I} - \mathbf{\Lambda}^{-1})_+^{1/2}. \quad (3.23)$$

The above equation is obtained by choosing the error weight matrix as $\mathbf{W}_e = \mathbf{\Lambda}$ in (3.18a). It is shown in [12] that the optimum precoder and combiner that maximize the information rate are given by (3.17a) and (3.17b) for this particular choice of error weights. In other words, the maximum information rate design is just a special case of the generalized WMMSE design. The choice of the error weights $\mathbf{W}_e = \mathbf{\Lambda}$ tells us that stronger eigen subchannels support higher rates compared to the weaker ones, and hence need to be more heavily weighted in the cost function. This corresponds with the waterfilling solution for optimal power allocation, where more power is allocated to stronger subchannels.

The maximum information rate design finds applications in adaptive modulation systems, where more power and higher order modulation are used on eigen subchannels with higher gains to improve the total data rate. The performance of these systems is evaluated in Section ??, based on simulation results for different system settings.

3.2.3 Quality of Service Design

Imagine an application where different types of information with different SNR requirements for successful transmission need to be simultaneously sent in different data streams. A concrete example is a multimedia application that transmits video and audio data in different data streams. For these Quality-of-Service (QoS) applications, it is necessary to design our system such that the different eigen subchannels have different SNRs.

First, we note that the expression for MSE matrix can be simplified by using the expressions (3.17a) and (3.17b) for the optimum precoder and combiner to

$$\text{MSE} = \mathbb{E}[\mathbf{e} \mathbf{e}^H] = \mathbf{W}_e^{-1/2} \mathbf{\Lambda}^{-1/2} \mu^{1/2}. \quad (3.24)$$

Let us now define the SNR matrix as

$$\text{SNR} = (\mathbf{G} \mathbf{H} \mathbf{F})^H (\mathbf{G} \mathbf{C}_n \mathbf{G})^{-1} (\mathbf{G} \mathbf{H} \mathbf{F}) \quad (3.25a)$$

$$= \mathbf{\Phi}_f^2 \mathbf{\Lambda} = (\mu^{-1/2} \mathbf{\Lambda}^{1/2} \mathbf{W}_e^{1/2} - \mathbf{I})_+ \quad (3.25b)$$

where we again used the expressions for the optimum precoder and combiner to simplify the first equality. It is easy to see that the SNR and the MSE are inversely correlated as $\text{SNR} = (\text{MSE}^{-1} - \mathbf{I})_+$. Hence equal SNRs on the eigen subchannels implies equal MSEs regardless of the choice of error weights.

We wonder how to choose the error weight matrix $\mathbf{W}_e$ to achieve any set of relative output SNRs across the eigen subchannels. The problem boils down to the following equation with respect to the weight matrix

$$d\mathbf{D} = (\mu^{-1/2} \mathbf{\Lambda}^{1/2} \mathbf{W}_e^{1/2} - \mathbf{I})_+ \quad (3.26)$$

where $\mathbf{D}$ is the diagonal matrix with the desired relative SNRs across the eigen subchannels and $d > 0$ is a scalar. The solution is given by [12]

$$\mathbf{W}_e^{1/2} = \mu^{1/2}(d\mathbf{D} + \mathbf{I})\mathbf{\Lambda}^{-1/2} \quad \text{with} \quad \mu^{-1/2} = \frac{\text{tr}(\mathbf{\Lambda}^{-1/2}\mathbf{W}_e^{1/2})}{P_t + \text{tr}(\mathbf{\Lambda}^{-1})} \quad \text{and} \quad d = \frac{P_t}{\text{tr}(\mathbf{D}\mathbf{\Lambda}^{-1})}, \quad (3.27)$$

where we assumed $\Phi_f > 0$ for simplicity.

Imagine a simple application that requires each data stream to be equally reliable. This means that all R_H eigen subchannels must have an equal error rate and hence equal output SNRs. The latter requirement can be satisfied trivially by selecting $\mathbf{D} = \mathbf{I}$. The resulting optimum precoder matrix $\mathbf{F}$ and combiner matrix $\mathbf{G}$ are given by [12]

$$\mathbf{F} = \mathbf{V}\Phi_f \quad \text{with} \quad \Phi_f = \sqrt{\frac{P_t}{\text{tr}(\mathbf{\Lambda}^{-1})}} \mathbf{\Lambda}^{-1/2} \quad (3.28a)$$

$$\mathbf{G} = \Phi_g \mathbf{V}^H \mathbf{H}^H \mathbf{C}_n^{-1} \quad \text{with} \quad \Phi_g = \frac{\sqrt{P_t \text{tr}(\mathbf{\Lambda}^{-1})}}{P_t + \text{tr}(\mathbf{\Lambda}^{-1})} \mathbf{\Lambda}^{-1/2} \quad (3.28b)$$

The power allocation policy for a QoS design with equal error rates assigns more power to the eigen subchannels with less channel gain while it allocates less power to the subchannels with stronger eigenmodes so that the output SNRs and MSEs of each subchannel are equal. Furthermore, no subchannels are dropped, regardless of the channel realization. It can be shown that the optimum precoder and combiner matrix transform the transfer matrix of the system into a scaled identity matrix.

Part II

Precoding for Satellite Communications

Part III

Appendices

Appendix A

Waterfilling for Power Allocation

In this appendix, we solve the problem of optimal power allocation across independent scalar Gaussian channels using the method of Lagrangian multipliers. The key principles of this method are explained alongside the derivation, because it is also employed to solve other optimization problems encountered in this dissertation. The solution to the optimal power allocation problem gives rise to the waterfilling algorithm, which is discussed at the end of this appendix.

A.1 Optimal Power Allocation

Consider N independent SISO Gaussian transmission channels. Optimal power allocation across these channels can be formulated as a constrained maximization problem. The objective is to distribute the total available transmit power among the different channels such that the combined achievable rate is maximized.

Consider the following general optimization problem:

$$\begin{aligned} \max_{\{p_n\}} \quad & \sum_{n=1}^N \alpha_n \log_2 (1 + \gamma_n p_n) \\ \text{s. t.} \quad & \sum_{n=1}^N p_n = P_t \quad \text{and} \quad p_n \geq 0 \quad \forall n = 1, \dots, N \end{aligned} \tag{A.1}$$

where p_n denotes the power allocated to channel n , and P_t denotes the total available transmit power. The coefficients α_n are non-negative weighting factors that allow different channels to be prioritized unequally, whereas $\gamma_n > 0$ denotes the ratio between the channel gain and the noise power of channel n .

To solve the constrained optimization problem in (A.1), the method of Lagrangian multipliers is employed. The core idea of this method is to convert a constrained optimization problem into an unconstrained one by absorbing the constraints into a modified objective function, known as the Lagrangian. This is achieved by introducing auxiliary variables, called Lagrange multipliers, one for each constraint, which penalize violations of the constraints.

Since the optimization problem in (A.1) involves both an equality constraint (total power budget) and inequality constraints (non-negativity of the individual power allocations), both types must be incorporated into the Lagrangian. Equality constraints enter with an unconstrained multiplier $\mu \in \mathbb{R}$, whereas inequality constraints of the form $p_n \geq 0$ are rewritten as $-p_n \leq 0$ and enter with a non-negative multiplier $\lambda_n \geq 0$, ensuring that the penalty is only active when the constraint is binding. The resulting Lagrangian is

$$\mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = \sum_{n=1}^N \alpha_n \log_2(1 + \gamma_n p_n) - \mu \left(\sum_{n=1}^N p_n - P_t \right) + \sum_{n=1}^N \lambda_n p_n \quad (\text{A.2})$$

where μ is the Lagrange multiplier associated with the total power constraint, and λ_n are the multipliers associated with the non-negativity constraints on the individual power allocations.

Since the objective function is concave and all constraints are affine, the optimization problem is convex. For convex problems, the Karush-Kuhn-Tucker (KKT) conditions are both necessary and sufficient for a solution to be globally optimal [13, 14]. These conditions are given by

$$1. \text{ Stationarity:} \quad \nabla_{\mathbf{p}} \mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = 0 \quad (\text{A.3a})$$

$$2. \text{ Primal feasibility:} \quad p_n \geq 0 \quad \text{and} \quad \sum_{n=1}^N p_n = P_t \quad (\text{A.3b})$$

$$3. \text{ Dual feasibility:} \quad \lambda_n \geq 0 \quad (\text{A.3c})$$

$$4. \text{ Complementary slackness:} \quad \begin{cases} p_n > 0 \implies \lambda_n = 0 \\ p_n = 0 \implies \lambda_n \geq 0 \end{cases} \quad (\text{A.3d})$$

From the KKT conditions, we can derive the optimal power allocation for each SISO channel. The derivation proceeds by successively applying the stationarity, complementary slackness, and primal feasibility conditions.

The stationarity condition requires that the gradient of the Lagrangian with respect to each optimization variable vanishes at the optimum. This expresses the balance between the marginal gain in the achievable rate and the penalties introduced by the power constraints. Applying this condition to the Lagrangian yields:

$$\nabla_{\mathbf{p}} \mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = 0 \quad \Longleftrightarrow \quad \frac{\partial \mathcal{L}}{\partial p_n} = \frac{\alpha_n \gamma_n}{\ln(2)(1 + \gamma_n p_n)} - \mu + \lambda_n = 0, \quad \forall n = 1, \dots, N \quad (\text{A.4})$$

The complementary slackness condition links the non-negativity constraint on the allocated power to its associated Lagrange multiplier. When a positive amount of power is allocated to channel n , the corresponding multiplier λ_n is zero, and the stationarity condition reduces to:

$$\frac{\alpha_n \gamma_n}{\ln(2)(1 + \gamma_n p_n)} - \mu = 0 \quad (\text{A.5a})$$

$$\Longleftrightarrow p_n = \frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \quad (\text{A.5b})$$

The primal feasibility condition enforces the non-negativity of the power allocations. As a result, the above expression for p_n must be clipped to zero whenever it becomes negative:

$$p_n = \left(\frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \right)_+ \quad (\text{A.6})$$

Finally, the primal feasibility condition also requires the total allocated power to equal the available transmit power P_t . By substituting the expression for p_n from (A.6) into the total power constraint, we obtain an implicit equation for the Lagrangian multiplier μ :

$$\sum_{n=1}^N \left(\frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \right)_+ = P_t \quad (\text{A.7})$$

Since μ cannot be obtained in closed form, it is determined iteratively such that the total allocated power satisfies (A.7). Owing to the monotonic relationship between μ and the total allocated power, a bisection method is typically employed to compute the optimal value of μ numerically.

A.2 Waterfilling Algorithm

A more efficient alternative to the bisection method exploits the geometric structure of (A.7). Associate with each channel n a container of width α_n and floor height $1/(\alpha_n \gamma_n)$. The total available transmit power P_t is represented as water that is poured simultaneously into all containers. Because all containers share a common water level, channels with a lower floor and a wider container receive more water. A container remains empty if its floor height lies above the common water level, meaning no power is allocated to that channel. The water volume in each channel container is exactly equal to the power allocated to that channel under the optimal solution [15]. This interpretation is illustrated in Fig. A.1 for a system with $N = 4$ channels.

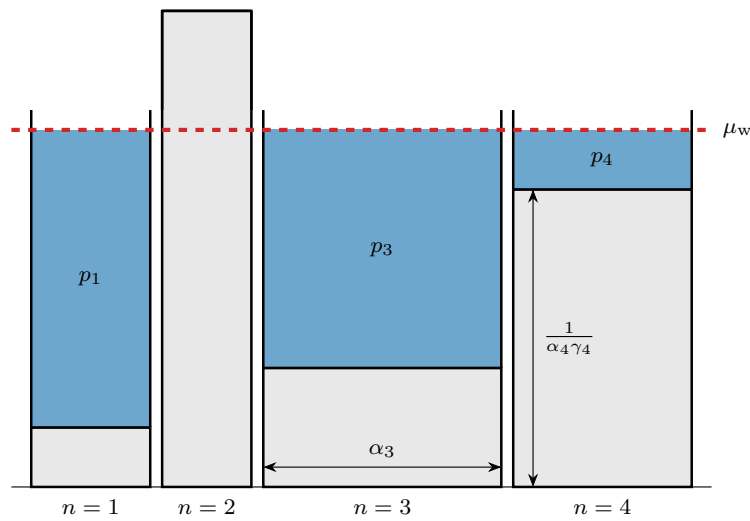


Figure A.1: A visualization of the waterfilling algorithm for optimal power allocation strategy in a system with $N = 4$ channels.

The complete procedure for determining the water level and the corresponding power allocation is summarized in Algorithm 1.

Algorithm 1 Waterfilling Algorithm for power allocation across Gaussian SISO subchannels

Require:

- The total transmit power: P_t ;
- The weight factor of each channel: α_n for $n = 1, \dots, N$;
- The ratios between channel gain and noise power: $\gamma_n = g_n^2/N_0$ for $n = 1, \dots, N$.

Ensure: The optimal power allocation p_n for $n = 1, \dots, N$.

Initialization: Determine the width and height of all channel containers.

Find the permutation $\mathcal{P} : n \rightarrow i$ between the original index n and the sorted index i that sorts the weighted ratios between channel gain and noise power $\alpha_n \gamma_n$ in decreasing order ($O(N \log N)$).

1: $w_i \leftarrow \alpha_{\mathcal{P}(i)}$, $h_i \leftarrow 1/(\alpha_{\mathcal{P}(i)} \gamma_{\mathcal{P}(i)})$

Iteration: Determine the optimal number of active channels ($O(\log N)$).

2: $N^{\text{LB}}, N^{\text{UB}} \leftarrow 0, N$

3: **while** $N^{\text{UB}} - N^{\text{LB}} > 1$ **do**

Update the number of active channels:

4: $N^{\text{iter}} \leftarrow \lfloor (N^{\text{UB}} + N^{\text{LB}})/2 \rfloor$

Compute the minimum power required to activate N^{iter} channels:

5: $P_{\min} \leftarrow \sum_{i=1}^{N^{\text{iter}}} w_i (h_{N^{\text{iter}}} - h_i)$

Update the bound on the number of active channels:

6: **if** $P_{\min} < P_t$ **then** $N^{\text{LB}} \leftarrow N^{\text{iter}}$ **else** $N^{\text{UB}} \leftarrow N^{\text{iter}}$

7: **end while**

Termination: Compute the power allocated to each channel.

Fill the N^{LB} active stream containers:

8: $p_i \leftarrow w_i (h_{N^{\text{LB}}} - h_i)$ for $i = 1, \dots, N^{\text{LB}}$

Distribute the remaining power among the active channels according to their weight:

9: $p_i \leftarrow p_i + (w_i / \sum_{i'=1}^{N^{\text{LB}}} w_{i'}) \cdot (P_t - \sum_{i'=1}^{N^{\text{LB}}} p_{i'})$ for $i = 1, \dots, N^{\text{LB}}$

Set the power of the inactive channels to zero:

10: $p_i \leftarrow 0$ for $i = N^{\text{LB}} + 1, \dots, N$

Use the inverse permutation $\mathcal{P}^{-1}$ to map the allocated power back to the original index n .

11: $p_n \leftarrow p_{\mathcal{P}^{-1}(i)}$

Return p_n for $n = 1, \dots, N$

A.3 Examples

Piecewise Frequency-Selective Channel

Consider a channel with a piecewise constant frequency response, partitioned into N non-overlapping subbands. Subband n spans the frequency interval $[B_{n-1}, B_n]$ and has a bandwidth $\beta_n = |B_n - B_{n-1}|$, a constant channel gain g_n , and experiences additive white Gaussian noise with power spectral density N_0 .

The capacity of this channel can then be written as

$$C = \max_{|S(f)|^2} \int_f \log_2 \left(1 + \frac{|H(f)|^2}{N_0} |S(f)|^2 \right) df \quad \text{s.t.} \quad \int_f |S(f)|^2 df \leq P_t \quad (\text{A.8a})$$

$$= \max_{|S(f)|^2} \sum_{n=1}^N \int_{B_{n-1}}^{B_n} \log_2 \left(1 + \frac{g_n}{N_0} |S(f)|^2 \right) df \quad \text{s.t.} \quad \int_f |S(f)|^2 df \leq P_t \quad (\text{A.8b})$$

$$= \max_{p_n} \sum_{n=1}^N \beta_n \log_2 \left(1 + \frac{g_n}{N_0} \frac{p_n}{\beta_n} \right) \quad \text{s.t.} \quad \sum_{n=1}^N p_n \leq P_t \quad (\text{A.8c})$$

where $S(f)$ denotes the power spectral density of the transmitted signal and $H(f)$ the frequency response of the channel. In the final step, Jensen's inequality and the concavity of the logarithm imply that the capacity-achieving power spectral density is constant within each subband. Accordingly, if p_n denotes the total power allocated to subband n , then $|S(f)|^2 = p_n/\beta_n$, for $f \in [B_{n-1}, B_n]$.

The optimization problem in (A.8c) is a special case of (A.1) with $\alpha_n = \beta_n$ and $\gamma_n = g_n/(N_0\beta_n)$. Hence, the capacity-achieving power allocation can be obtained by applying the waterfilling algorithm described in the previous section.

Single-User MIMO Channel

As shown in Section 3.1.1, a SU-MIMO channel can be decomposed into R_H independent parallel SISO eigen subchannels by means of the SVD of the channel matrix.

The capacity optimization problem takes the form

$$C = \max_{\{p_n\}} \sum_{n=1}^{N_s} \log_2 \left(1 + \frac{\sigma_n}{N_0} p_n \right) \quad \text{s.t.} \quad \sum_{n=1}^{N_s} p_n \leq P_t,$$

where p_n is the power allocated to the n -th eigen subchannel with channel gain σ_n and N_0 is the noise power on each receive antenna.

This is a special case of (A.1) with $\alpha_n = 1$ and $\gamma_n = \sigma_n/N_0$. Hence, the optimal power allocation, in the sense that it achieves the capacity, can be obtained by applying the waterfilling algorithm described in the previous section.

Appendix B

Calculations

In this appendix, we provide the calculations that are relevant to the main text but were omitted for brevity. The appendix is organized into separate sections that correspond to specific intermediate results used throughout the dissertation.

Total Transmit Power Constraint

The following calculations derive the total transmit power as a function of the precoding matrix for both the SU-MIMO and MU-MIMO system models. In both cases, the power allocation is assumed to be incorporated into the precoding matrix, and the data symbols are assumed to be zero-mean, mutually uncorrelated, and normalized, i.e. $\mathbb{E}[\mathbf{a}\mathbf{a}^H] = \mathbf{I}_{N_S}$.

Consider the SU-MIMO system model as described in Section 2.2.1. The total transmit power is constraint is given by:

$$\mathbb{E}[\|\mathbf{x}\|^2] = \text{tr}(\mathbf{F}^H \mathbf{F}) \leq P_t.$$

Calculation:

$$\begin{aligned} P_t &\geq \mathbb{E}[\|\mathbf{x}\|^2] = \mathbb{E}[\mathbf{x}^H \mathbf{x}] = \mathbb{E}[(\mathbf{F}\mathbf{a})^H (\mathbf{F}\mathbf{a})] = \mathbb{E}[\mathbf{a}^H \mathbf{F}^H \mathbf{F} \mathbf{a}] \\ &= \mathbb{E}[\text{tr}(\mathbf{a}^H \mathbf{F}^H \mathbf{F} \mathbf{a})] = \mathbb{E}[\text{tr}(\mathbf{F}^H \mathbf{F} \mathbf{a} \mathbf{a}^H)] \\ &= \text{tr}(\mathbb{E}[\mathbf{F}^H \mathbf{F} \mathbf{a} \mathbf{a}^H]) = \text{tr}(\mathbf{F}^H \mathbf{F} \mathbb{E}[\mathbf{a} \mathbf{a}^H]) \\ &= \text{tr}(\mathbf{F}^H \mathbf{F} \mathbf{I}_{N_S}) = \text{tr}(\mathbf{F}^H \mathbf{F}) \end{aligned}$$

Consider the MU-MIMO system model as described in Section 2.2.2. The total transmit power is constraint is given by:

$$\mathbb{E}[\|\mathbf{x}\|^2] = \sum_{k=1}^K \text{tr}(\mathbf{F}_k^H \mathbf{F}_k) \leq P_t.$$

Calculation:

$$\begin{aligned}
P_t &\geq \mathbb{E}[\|\mathbf{x}\|^2] = \mathbb{E}[\mathbf{x}^H \mathbf{x}] = \mathbb{E}\left[\left(\sum_{k=1}^K \mathbf{F}_k \mathbf{a}_k\right)^H \left(\sum_{k'=1}^K \mathbf{F}_{k'} \mathbf{a}_{k'}\right)\right] \\
&= \mathbb{E}\left[\sum_{k=1}^K \sum_{k'=1}^K (\mathbf{F}_k \mathbf{a}_k)^H \mathbf{F}_{k'} \mathbf{a}_{k'}\right] = \mathbb{E}\left[\sum_{k=1}^K \sum_{k'=1}^K \mathbf{a}_k^H \mathbf{F}_k^H \mathbf{F}_{k'} \mathbf{a}_{k'}\right] \\
&= \mathbb{E}\left[\sum_{k=1}^K \sum_{k'=1}^K \text{Tr}(\mathbf{a}_k^H \mathbf{F}_k^H \mathbf{F}_{k'} \mathbf{a}_{k'})\right] = \mathbb{E}\left[\sum_{k=1}^K \sum_{k'=1}^K \text{Tr}(\mathbf{F}_k^H \mathbf{F}_{k'} \mathbf{a}_{k'} \mathbf{a}_k^H)\right] \\
&= \sum_{k=1}^K \sum_{k'=1}^K \text{Tr}(\mathbf{F}_k^H \mathbf{F}_{k'} \mathbb{E}[\mathbf{a}_{k'} \mathbf{a}_k^H]) \\
&= \sum_{k=1}^K \sum_{k'=1}^K \text{Tr}(\mathbf{F}_k^H \mathbf{F}_{k'} \mathbf{I}_{N_S} \delta_{kk'}) \\
&= \sum_{k=1}^K \text{Tr}(\mathbf{F}_k^H \mathbf{F}_k)
\end{aligned}$$

Equivalent SU-MIMO Channel Representation

Applying the singular value decomposition of the channel matrix together with a change of variables yields an equivalent representation of the SU-MIMO channel. The following calculation shows how the original channel input-output relation is transformed into a set of parallel scalar Gaussian subchannels.

Calculation:

$$\begin{aligned}
\mathbf{y} &= \mathbf{H}\mathbf{x} + \mathbf{n} \\
\Leftrightarrow \mathbf{y} &= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \mathbf{x} + \mathbf{n} \\
\Leftrightarrow \mathbf{U}\tilde{\mathbf{y}} &= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \mathbf{V}\tilde{\mathbf{x}} + \mathbf{n} \\
\Leftrightarrow \tilde{\mathbf{y}} &= \mathbf{U}^H \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \mathbf{V}\tilde{\mathbf{x}} + \mathbf{U}^H \mathbf{n} \\
\Leftrightarrow \tilde{\mathbf{y}} &= \mathbf{\Sigma}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}.
\end{aligned}$$

Mutual Information under Optimal Precoder and Combiner

The following calculation verifies that the MI between the transmitted data symbols $\mathbf{a}$ and the decision variables $\mathbf{z}$ equals the MI between the channel input $\mathbf{x}$ and the channel output $\mathbf{y}$ when the optimal precoder and combiner from (3.8) are employed. The result is obtained by substituting $\mathbf{F} = \mathbf{V}_{N_s} \sqrt{\mathbf{P}}$ and $\mathbf{G} = \mathbf{U}_{N_s}^H$ into (3.7b) and replacing the channel matrix by its SVD (3.1).

Calculation:

$$\begin{aligned}
I(\mathbf{a}; \mathbf{z}) &= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} (\mathbf{U}_{N_s}^H \mathbf{U}_{N_s})^{-1} (\mathbf{U}_{N_s}^H (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^H) \mathbf{V}_{N_s} \sqrt{\mathbf{P}}) (\mathbf{U}_{N_s}^H (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^H) \mathbf{V}_{N_s} \sqrt{\mathbf{P}})^H \right) \right) \\
&= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} \mathbf{I}_{N_s} (\mathbf{U}_{N_s}^H \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H \mathbf{V}_{N_s} \sqrt{\mathbf{P}}) (\mathbf{U}_{N_s}^H \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H \mathbf{V}_{N_s} \sqrt{\mathbf{P}})^H \right) \right) \\
&= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} \left(\begin{bmatrix} \mathbf{I}_{N_s} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{\Sigma}_{R_H} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{I}_{N_s} \\ \mathbf{0} \end{bmatrix} \sqrt{\mathbf{P}} \right) \left(\begin{bmatrix} \mathbf{I}_{N_s} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{\Sigma}_{R_H} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{I}_{N_s} \\ \mathbf{0} \end{bmatrix} \sqrt{\mathbf{P}} \right)^H \right) \right) \\
&= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} (\mathbf{\Sigma}_{N_s} \sqrt{\mathbf{P}}) (\mathbf{\Sigma}_{N_s} \sqrt{\mathbf{P}})^H \right) \right) \\
&= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} \mathbf{\Sigma}_{N_s}^2 \mathbf{P} \right) \right) \\
&= \log_2 \left(\prod_{i=1}^{N_s} \left(1 + \frac{\sigma_i^2}{N_0} p_i \right) \right) \\
&= \sum_{i=1}^{N_s} \log_2 \left(1 + \frac{\sigma_i^2}{N_0} p_i \right)
\end{aligned}$$

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